

BINAY - Lighting Up The Future



BINAY LED-based HIGH, MEDIUM and LOW Intensity Aviation Obstruction Light Beacons

As per International Civil Aviation Organisation (ICAO) requirements Available in Low Intensity, Medium Intensity and High Intensity versions (as per International Civil Aviation Organization guidelines), BINAY's patented LED Aviation Lights come with 5-year/3-year warranties

LED Obstruction Lighting for:

- Industrial chimneys and smokestacks
- Transmission, microwave and cellular towers
- Radio, TV and similar structural towers
- High-rise buildings and structures
- Airports and airfields

The BINAY LED Aviation Obstruction Light offers the following advantages:

- Fit-and-forget maintenance-free operation
- A long life of 100,000 hours (20 years at 12 hours daily burning)
- Pays for itself within a short period of operation in the form of reduced installation, maintenance and servicing costs
- Quick Installation; Reliable operation 365 days per year
- Shock-proof and vibration-resistant
- Over-Designed Intensity to allow for natural LED Intensity degradation over its operating lifetime



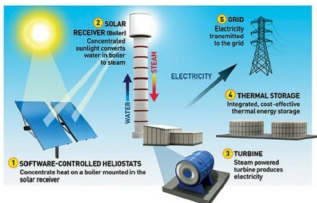
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POWERING LED TECHNOLOGIES WORLDWIDE SINCE 1983

Hydricity, the breakthrough in solar

Scientists at Purdue University and Federal Polytechnic School of Lausanne have developed Hydricity, which combines solar thermal plants with hydrogen production technology to synthesise energy at a greater scale and escalated efficiency.



A conventional solar thermal plant (Image courtesy: www.kimroybailey.com)

In this technique, a hybrid system uses solar energy to simultaneously generate steam by heating water, as well as hydrogen by electrolysis. Steam generated is used to run turbines as in a conventional solar thermal power plant, and hydrogen produced is stored away for future use.

Solar power is mainly harnessed by using solar power cells that convert light into electricity, or solar thermal power plants, which use the Sun's heat to generate steam, which is then used to run turbines to produce electricity. Both these processes are only fairly efficient, with an average efficiency of around 30 per cent. Hydricity is an attempt to efficiently harvest solar power that could meet the world's need for energy.

According to the team, the overall efficiency of Hydricity is 35 per cent, with 50 per cent efficiency in hydrogen production and 46 per cent efficiency in immediate electricity generation. Boost from the average 30 per cent is achieved by a series of high-pressure and low-pressure turbines, connected successively.

Electrical circuits inside living plants developed

Researchers at Linköping University, Sweden, have created analogue and digital electronics circuits inside living plants. The group at Laboratory of Organic Electronics (LOE), led by Prof. Magnus Berggren, has used the vascular system of living roses to build key components of electronic circuits.

Plants are complex organisms that rely on the transport of ionic signals and hormones to perform necessary functions. But, these operate on a very slow time scale, making